

AI Voice Receptionist – Architecture

Using Twilio, n8n, LLMs, and Google Sheets

Overview

This document explains the architecture of an AI Voice Receptionist system built using Twilio for telephony, n8n for workflow automation, Large Language Models (LLMs) for reasoning, and Google Sheets for call logging and lead storage.

High-Level Architecture Flow

Caller → Twilio Voice Webhook → n8n Workflow → Speech-to-Text → LLM Processing → Text-to-Speech → Twilio Voice Response → Caller

Call metadata and captured information such as caller number, intent, and summary are stored in Google Sheets for record keeping and follow-up.

Core Components

- Twilio Voice: Receives and manages inbound phone calls and forwards them to a webhook.
- n8n Workflow: Orchestrates the entire call flow and integrates all services.
- Speech-to-Text (STT): Converts caller voice into text for AI understanding.
- LLM (AI Model): Analyzes caller intent and generates appropriate responses.
- Text-to-Speech (TTS): Converts AI-generated responses back into voice.
- Google Sheets: Stores call logs, caller details, and conversation summaries.

Step-by-Step Call Flow

- 1 A caller dials the business phone number.
- 2 Twilio receives the incoming call and triggers a webhook.
- 3 The webhook sends call data to an n8n workflow.
- 4 Caller speech is converted into text using Speech-to-Text.
- 5 The transcribed text is sent to the LLM for intent analysis.
- 6 The AI generates a professional response.
- 7 The response text is converted into speech using Text-to-Speech.
- 8 Twilio plays the AI-generated voice back to the caller.
- 9 Call details and summaries are saved to Google Sheets.

Business Use Cases

- AI receptionist for offices and clinics
- Automated customer support calls
- Service inquiry and information hotlines
- Lead capture and call logging
- Virtual IVR replacement

Resume-Ready Architecture Summary

Designed and implemented an AI voice receptionist architecture using Twilio, n8n, speech processing, and large language models with automated call logging in Google Sheets.